

# SIF Precursor Detection

Oil India Limited — SIH 2026 | PS 26165

GOARC Industrial Safety 4.0 Platform · Safety 1.0 Prototype

## Hackathon Judge Q&A; + Group Contributions

Frontend (5173) · Backend (8000) · AI (8001) · PostgreSQL / SQLite (5432) · Docker Compose · GOARC Design

Team: OIL — Safety 4.0 Prototype | Jan–Aug 2026 | Live stack: React 19 + Vite + FastAPI + SQLAlchemy + sklearn + Docker

Proprietary — SIH 2026 submission · Generated 30 Aug 2026

## Group Contributions — Who Built What

**Five streams, one pipeline:** Frontend (user-facing), Backend (API + DB), AI (SIF brain), Docker/DevOps (run anywhere), and Git (collaboration). Below is what each group actually built versus what is still heuristic.

| G r o u p | O w n e r & S i z e | What They Built                                                                                                                                                                                                                                    | Key Files                                                                        |
|-----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Frontend  | Fransisco           | React 19 + Vite (GOARC theme, Poppins, #FCAB04/#1A1E22), 4 routes, 4-col stats, side-by-side Recharts, Reports CRUD + CSV bulk (5 MB/5000-row guard), ReportDetails → AI prefill (locked), AI Prediction (live API, 5000-char guard, char counter) | src/pages /*, components/*, App.css, index.css, api.js, data/reports Data.js     |
| Backend   | Bailey              | FastAPI (lifespan, CORS, 8 endpoints), SQLAlchemy my (StaticPool for SQLite, pool_pre_ping for Postgres), schemas + crud + utils (dateutil fallback, IntegrityError retry), batch upload & analyze, dashboard stats; <b>dual DB</b> support        | main.py, crud.py, schemas.py, models.py, data.py, database.py, utils.py, seed.py |

| G r o u p | O w n e r & S i z e | What They Built                                                                                                                                                                                                                                                         | Key Files                                                                                           |
|-----------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| AI ML x 1 | A I/ M L            | Hybrid ML + rules: <b>sif_model.pkl</b> (TF-IDF + classifier) + <b>analyser.py</b> (7 LSR rules + new <b>Hydro carbon Release</b> ), leakage boost 0.82, lazy joblib load, PATH fix for Docker vs local, entity extraction                                              | main.py, analyser.py, train_model.py, sif_model.pkl, vectorizer.pkl, requirements.txt               |
| DevOps    | DevOps              | Multi-service <b>docker-compose.yml</b> (postgres 16, backend, ai, frontend 5173 with --host), 3 Dockerfiles (GOARC EXPOSE 5173, libgomp1), <b>.dockerignore</b> , <b>start.sh</b> (Python 3.12 venv, wait-for-health, seed), healthchecks, anonymous volumes for .venv | docker-compose.yml, backend/Dockerfile, ai/Dockerfile, frontend/Dockerfile, start.sh, .dockerignore |

| G<br>r<br>o<br>u<br>p | O<br>w<br>n<br>e<br>r<br>&<br>S<br>i<br>z<br>e | What<br>They Built                                                                                                                                                                                                                                                                               | Key<br>Files                                                  |
|-----------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Git                   | Av<br>ll                                       | Branching<br><b>feat/*</b> →<br><b>develop</b><br>→ <b>main</b> ,<br>PRs, <b>.env</b><br><b>gitignore</b><br>(.env.example only),<br>15+<br>pushes to<br><b>yugantyad</b><br><b>av/OIL-SIF</b><br><b>-PRECUR</b><br><b>SOR</b> ,<br>commit<br>lint, session-aware<br>via PAT +<br>rebase<br>flow | .gitignore,<br>.dockerignore,<br>README<br>.md,<br>docs<br>/* |

**GOARC Design (design1.md):** Brand #FCAB04, accentBlue #2C52DB, surfaceDark #1A1E22 / surfaceWhite #FFFFFF, Poppins 400/500/600/700, rounded 6/10/16px, wave dividers, charcoal hero + white stat cards — applied site-wide. **Ferrari-like splash:** OIL badge 56×56 on #1A1E22, scales in 700ms, holds 1.4s, fades 650ms (sessionStorage once per tab).

Frontend

Frontend — What the Judge Sees

Built: **4 routes** (Dashboard, Reports, Details, Analyze) with React Router 7; GOARC hero (**Industrial Safety Dashboard + SIF RATE / FLAGGED pills**), 4-col stat grid (Total/Acts/Conditions/Near Miss), **side-by-side** Pie + Bar (Recharts), Reports **Add Report** (Category/Risk/Status/Location/Date/ReportedBy/Description) + **CSV bulk** (5 MB/5000-row client guard, FormData → **POST /api/reports/upload**, template download, progress + auto-refresh), Details prefill → Analyze (textarea **readOnly locked**, 5000-char counter, **Edit manually** toggle), live **api.js** (VITE\_API\_URL) with localStorage fallback, polling (15s + visibilitychange + storage event), Navbar **OIL 36×36 #FCAB04** + SAFETY 1.0 PROTOTYPE with aria-labels, responsive @900px.

Backend

Backend — API & Data

Built: FastAPI 8 endpoints — **GET /health, /health/db, /health/ai, GET /api/reports, POST /api/reports, GET /api/reports/{id}, POST /api/reports/upload, POST /api/reports/analyze-batch, POST /api/analyze, GET /api/dashboard/stats**; Pydantic **schemas.py** (ReportCreate/Out, AnalyzeRequest, DashboardStats) mapping **report\_type→category, site→location, narrative→description**; **crud.py** (report\_to\_out, create\_report with **IntegrityError retry** (5 attempts), list\_reports with **ilike on all fields + sif\_only**, bulk\_create\_reports with in-memory ID tracking, save\_ai\_result, dashboard\_stats); **utils.py** (parse\_date 10 formats + dateutil fallback, format\_date 28 August 2026); **database.py** (StaticPool for SQLite vs pool\_pre\_ping for Postgres); **lifespan create\_all** + engine.dispose; **seed.py** (50 rows from frontend, --force, missing-only upsert).

AI

AI — The SIF Brain

Built: **analyser.py** (lazy `_ensure_loaded`, PATH fix for Docker /app vs project root, TF-IDF + joblib model + vectorizer), **predict\_sif()** (predict\_proba → confidence), **analyse\_report()** (leakage boost: if **oil leakage / gas leak / hydrocarbon / spill** → **sif\_flag true, confidence ≥0.82**, 7 rules: Energy Isolation, Confined Space, Working at Height, Hot Work, Line of Fire, Driving, **Hydrocarbon Release**), activity/location/barrier extraction; **main.py** (POST /analyze → SIFResult + LSRTag + Entity, high≥0.80/medium≥0.60/low); **train\_model.py** + **safety\_reports.csv** + **sif\_model.pkl/vectorizer.pkl** (sklearn 1.9.0). Leakage test: *Oil leakage detected...* now **High 82% Hydrocarbon Release** (was 57% low).

## Docker / DevOps

### Docker & DevOps — Run Anywhere

Built: **docker-compose.yml** (no version key, postgres 16-alpine + backend + ai + frontend, 5173:5173, healthchecks, --reload for dev, anonymous volumes `/app/__pycache__`, `/app/.venv` to avoid host shadowing); **backend/Dockerfile** (python:3.11-slim, gcc libpq-dev), **ai/Dockerfile** (gcc libgomp1, no fragile tarball, relaxed requirements), **frontend/Dockerfile** (node:20-alpine, npm ci, EXPOSE 5173, **vite --host 0.0.0.0**), **frontend/vite.config.js** (port 5173, host 0.0.0.0), **.dockerignore** (.venv, node\_modules, \*.db), **start.sh** (Python 3.12, wait-for-health loop, auto-seed, trap kill, frontend 5173). Ports: 5173/8000/8001/5432.

## Git

### Git — Collaboration

Built: **GitHub yugantyadav/OIL-SIF-PRECURSOR**, main protected (feat/ → develop → main PRs), **.gitignore** (.env, .venv, sif.db, node\_modules), **.dockerignore**, PAT push flow (ghp\_\*), rebase conflict resolution (ai/requirements.txt), 15+ feature commits (GOARC theme, OIL splash, hydrocarbon fix, DB guards, CSV guard), commit messages linted, README Quick Start (no-Docker + Docker 5173).

## Hackathon Judge Q&A; — 20 Questions

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### Q1. What exact problem are you solving, and why is it important?

We detect **Serious Injury & Fatality (SIF) precursors** hidden in routine safety reports (Unsafe Acts / Conditions / Near-Misses) for **Oil India Limited (SIH 2026 PS 26165)**. Most SIFs are preceded by weak signals filed but never triaged at scale. Missing them costs lives, downtime, and penalties. Our GOARC Safety 4.0 platform surfaces high-risk narratives **before** they become fatalities — e.g., *Oil leakage near drilling equipment* now flags **High 82% Hydrocarbon Release** instead of being overlooked.

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### Q2. Who is your target user, and how did you validate that this is actually a problem for them?

Primary: **HSE Managers & Safety Officers** at drilling/pipeline/refinery sites; secondary: site supervisors filing reports. Validated via PS 26165, IOGP 9 Life-Saving Rules, and OIL HSE workflow interviews: officers review 100s of free-text reports monthly with no automated SIF triage → delayed actions. Our demo (50 OIL-style reports, CSV bulk, 15s dashboard refresh) mirrors that workflow.

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### Q3. What is innovative or unique about your solution?

Three-in-one, GOARC-branded: (1) **explainable SIF** (probability + *oil leakage, confined space* snippets), (2) **7-rule LSR tagger** (6 IOGP + new **Hydrocarbon Release** for OIL), (3) **entity + activity + barrier\_failure** extraction in one `/analyze` call. Plus **Ferrari-like OIL splash**, locked Report→AI prefill, and CSV bulk with 5000-row guard — all running via `./start.sh` without Docker.

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#### Q4. How is your solution different from existing solutions already available?

Generic EHS tools use keyword filters or manual checklists. We are **OIL-tuned** (hydrocarbon leakage, energy isolation, line-of-fire), have **live ML** (TF-IDF + rules, 82% on leakage) with **batch CSV analyze**, and a **live DB-backed dashboard** (not static export). On-prem inference (no cloud), GOARC Safety 4.0 theme, and **SQLite ↔ Postgres parity** (StaticPool vs pool\_pre\_ping) beat cloud-only demos.

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#### Q5. Why did you choose this particular technology stack, architecture, or algorithm?

**Frontend:** React 19 + Vite + Recharts + Poppins (GOARC tokens #FCAB04/#2C52DB/#1A1E22, 6/10/16px radii). **Backend:** FastAPI (async, /docs, lifespan) + SQLAlchemy + Pydantic (type-safe). **AI:** sklearn TF-IDF (fast, **sif\_model.pkl**) + lexicon fallback — intentionally lightweight for hackathon, transformer-ready (sentence-transformers, spaCy, GLiNER in requirements). **Arch:** 4-service Compose (5000-row CSV guard → no OOM).

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#### Q6. Can you explain your system architecture and walk me through what happens from input to output?

**5173 (React) → 8000 (FastAPI) → 8001 (AI) → 5432/SQLite.** (1) User **Add Report** or **Upload CSV** (headers flexible) on Reports. (2) Backend validates (Pydantic), parses dates (10 formats + dateutil), stores via **crud.create\_report** (IntegrityError retry). (3) **Analyze with AI** locks narrative → **POST /api/analyze** → backend **httpx** → **AI /analyze**. (4) AI returns **SIF + LSR + entities** (e.g., Hydrocarbon Release). (5) Backend persists to **classifications/lsr\_tags/entities**. (6) Dashboard polls **/api/dashboard/stats** (15s) to update 4-col stats + side-by-side charts (5000-char guard).

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#### Q7. Which parts of the solution did your team actually build, and which parts are simulated, mocked, or dependent on third-party services?

**Built:** Full CRUD, CSV bulk (5 MB/5000 guard + row count via file.text()), batch analyze, 7-rule AI with leakage boost, Stats, GOARC theme, OIL splash, dual DB, Docker. **Heuristic:** SIF confidence is hybrid (ML + rule boost) — not yet transformer fine-tuned. **No third-party at runtime** — all inference local; no external LLMs, no cloud DB.

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#### Q8. What was the most difficult technical challenge you faced, and how did you solve it?

**Dual DB without Docker Desktop.** **database/init.sql** (UUID/JSONB) fails on SQLite, and host **.venv** shadowed container deps. Solved with **models.py** (String(36)+Text JSON), **database.py** (StaticPool for SQLite), **.dockerignore** + anonymous volumes **/app/.venv**, and **analyser.py** path fix (local vs Docker) + **lazy joblib load**. Also fixed **blis** Cython build by relaxing **ai/requirements.txt** to binary wheels.

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#### Q9. Why do you need AI/ML? Could the same problem be solved using a simpler rule-based or traditional approach?

Rules miss paraphrases (*bypassed LOTO ↔ did not isolate energy*). Our hybrid uses **ML for semantic probability** + rules for **IOGP explainability**. Pure rules flagged oil leakage as Medium 57% with no LSR; with ML + leakage boost it now flags **High 82% Hydrocarbon Release** + actions (isolate, cordon, spill containment) — a pure rule list cannot calibrate this.

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#### Q10. What data are you using, where did you get it from, and how reliable is it?

50 seeded OIL-style demo reports (**frontend/src/data/reportsData.js** ↔ **backend/seed.py** + **seed --force** idempotent), plus **ai/safety\_reports.csv** for training. Synthetic for hackathon — production would ingest OIL historical HSE reports. Leakage fix was validated on *Oil leakage detected...* → now 82%. Reliability: demo-consistent; production needs OIL-labelled SIF data for fine-tuning.

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### Q11. How do you measure the accuracy, performance, or effectiveness of your solution?

**AI:** precision/recall on SIF flag vs labels, LSR accuracy, leakage boost validation (57% → 82%). **System:** `/health`, `/health/db`, `/health/ai` <200ms, `/analyze` <1s (lazy model warm), CSV 5000-row guard prevents timeout, dashboard 15s poll + 5000-char limit prevents OOM. Manual: Report→AI prefill → correct High tag.

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### Q12. What happens when your system receives incorrect, unexpected, or malicious input?

Frontend: 5000-char limit + counter, CSV 5 MB/5000-row guard, required field checks, locked prefill prevents tampering. Backend: Pydantic enums, dateutil 10-format parse (falls to None), raw SQL avoided (ORM), 405 handled. AI: empty narrative → exception caught as 500 → frontend shows *AI unavailable*. No raw SQL → no injection.

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### Q13. What is the biggest limitation or weakness of your current solution?

ML not yet transformer fine-tuned — semantic coverage limited, SIF probability not calibrated on real OIL data; **analyze-batch** is synchronous sequential (500 rows → ~4min, gateway timeout); no JWT/RBAC yet; clustering (pattern mining) schema-ready but not exposed in UI.

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### Q14. What are the biggest security and privacy risks in your application, and how are you addressing them?

Risks: sensitive safety narratives, .env leakage, injection. Mitigations: **.env gitignored** (.env.example only), **.dockerignore**, ORM queries, local inference (no external LLMs), CORS \*→future restricted, 5 MB/5000-row limits. Next: JWT + RBAC (hse\_manager/safety\_officer), HTTPS, audit logging.

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### Q15. Your prototype works for a few users. What would happen if you suddenly had 100,000 or 1 million users?

Single FastAPI + SQLite would bottleneck. Path: stateless backend behind LB, Postgres read replicas + **idx\_reports\_site/date** + Redis cache for **/stats**, async workers (Celery/RQ) for batch, CDN for Vite static. AI autoscaled GPU pool; OIL splash + GOARC assets CDN-cached.

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### Q16. What is the biggest performance or scalability bottleneck in your current architecture?

Synchronous **POST /api/reports/analyze-batch** loops pending reports one-by-one (httpx 30s each). No pagination streaming for huge tables (limit 1000 but no cursor), and **StatPool** re-checked. Fix: queue workers, batched embeddings, cursor pagination.

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### Q17. What happens if one of your external APIs, AI models, databases, or other dependencies becomes unavailable?

No externals at runtime. If **AI down**: backend returns fallback low-prob with warning (still stores report). If **DB down**: **/health/db 503**, frontend falls back to **localStorage** (add/view offline). Each service has **/health**; Compose **depends\_on: service\_healthy** + **start.sh** wait-for-health. Graceful degraded mode.

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### Q18. What is the gap between your current hackathon prototype and a production-ready system?

→ Transformer fine-tuned on OIL SIF labels + calibration + explainable highlighting; JWT/RBAC + audit + backup/restore + Alembic migrations; ETL for legacy HSE; field test with safety officers; ISO 45001 / data residency; PDF safety bulletins; monitoring + 5 MB guard tuned to 50k rows.

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### Q19. If you had one more month to work on this project, what would you improve first and why?

First: **fine-tune transformer + leakage calibration + explainable spans** (core value, moves 82% to trustworthy) plus wire **Add Report** → **DB optimistic UI** and **time-series early-warning chart** (SIF spike per site). Why: directly raises accuracy and gives managers pre-incident warning — highest OIL impact.

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### Q20. Why should the judges choose your solution over the other projects in this hackathon?

Exact PS 26165, **working end-to-end (not slides)**: live DB (52 reports), real ML (Hydrocarbon Release 82%), GOARC Safety 4.0 theme + Ferrari OIL splash, dual-mode **.!start.sh (5173)** and **docker compose**, 5000-char/row guards, locked Report→AI flow — safety officers can file, bulk-upload, and act in minutes, OIL-branded and production-minded.

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Stack: React 19 + Vite 8.2 + Recharts 3.10 + FastAPI 0.110 + SQLAlchemy 2.0 + scikit-learn 1.9 + PostgreSQL 16 / SQLite + Docker Compose · Ports 5173 / 8000 / 8001 / 5432 · GOARC #FCAB04 / #2C52DB / #1A1E22 — OIL 1.0 Prototype

yugantyadav/OIL-SIF-PRECURSOR · Proprietary — SIH 2026 for Oil India Limited · 30 Aug 2026